

Forecasting Electricity Prices

NORLYS ENERGY TRADING

Primer

The Danish power grid consists of two sub-grids called **DK1** (Western Denmark) and **DK2** (Eastern Denmark). Electricity flows between these grids and also with other neighboring countries (like Germany). The price of power is determined by supply and demand, and at 12:45 CET/CEST each day, the price of electricity for each hour of the following day is announced. We refer to these prices as **spot prices**. Accurate predictions of spot prices are useful for several purposes, e.g.,:

- secure the electricity price towards consumers;
- lock the electricity price when electricity is used in production;
- financial speculation.

The goal of this task is to predict spot prices in **DK1** before they are announced by exploiting various other datasets (actual and forecast) which are known to affect the spot price.

Data

The data consists of two `csv` files, `data_actual.csv` and `data_forecast.csv` respectively. The former is the actual (“true”) data which can be used to train models and validate predictions; the latter is the forecast data which can be used to train models and as predictors to forecast future spot prices. The data comprises:

- `valuetime` (timestamp): the UTC timestamp for which the corresponding value is valid (either forecasted or measured).
- `wind` (MW): wind power production in **DK1**.
- `solar` (MW): solar power production in **DK1**.
- `load` (MW): electricity consumption in **DK1**.
- `flow` (MW): import (> 0) / export (< 0) of electricity between **DK1** and Germany.
- `spot` (EUR/MWh): the spot price.

Actual Data

Shape: (2880, 6)

	valuetime	wind	solar	load	spot	flow
0	2020-10-31 23:00:00	2337.350098	0.0	1893.0	4.57	1300.0
1	2020-11-01 00:00:00	2179.510010	0.0	1837.0	4.28	1300.0
2	2020-11-01 01:00:00	2026.589966	0.0	1841.0	3.51	1300.0

Forecast Data

Shape: (2880, 5)

	valuetime	wind	solar	load	flow
0	2020-10-31 23:00:00	2980.510010	0.0	1905.119995	1300.0
1	2020-11-01 00:00:00	2637.489990	0.0	1821.660034	1300.0
2	2020-11-01 01:00:00	2222.360107	0.0	1768.150024	1300.0

Challenge description

- You should predict the hourly spot prices in **DK1** at no later than **09:00 UTC** for the following day, using the data described above and/or any other derived features that you may wish to consider (e.g., seasonalities).
- For such a ‘day-ahead’ forecast, this means we can use all actuals and forecasts up to UTC time 09:00 a particular day to train a model, call it M (the target is **spot**). Given M , we can then use the day-ahead forecast data in `data_forecast.csv` at the valuetimes for the following day to make predictions.
- Generate a series of spot price forecasts and evaluate your predictions using some traditional error metrics of your choice (e.g., MAPE, RMSE, bias).

Consider this to be a one work-day assignment so please match ambitions accordingly. You may wish to do some exploratory data analysis prior to building and testing a model. The idea is simply to have some material we can discuss—not to have a state-of-the-art forecast!